AI-Driven Urban US Housing Insights for Sustainable City Living

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Chapter 1

Introduction

1.1 Narrowing the Scope: From SDGs to Urban Housing

In the previous semester, my research focused on the broader landscape of sustainable development, exploring the interconnected nature of the Sustainable Development Goals (SDGs) through a data-driven framework. While this approach revealed important systemic relationships, its wide scope posed certain limitations in terms of practical application and contextual depth.

To address this, the focus of this semester's project was narrowed to a specific and highly relevant domain analyzing of sustainable housing market in USA. Housing operates at the intersection of economic, social, and environmental dimensions, making it a valuable lens for examining sustainability in a localized and measurable way. It not only reflects broader systemic challenges such as inequality, affordability, and urban planning but also directly impacts the livability and resilience of cities.

While the first semester concentrated on exploring SDGs from a global perspective through a literature-based methodology, this phase leverages real-world data to identify patterns and explore housing market behavior using data analytics techniques. By shifting from a macro-level theoretical framework to a focused, applied analysis, this project aims to translate conceptual insights into actionable understanding bridging the gap between global sustainability goals and localized urban realities.

1.2 Housing, Urbanization, and the Sustainable Cities Agenda

In recent years, rapid urbanization and rising housing prices have significantly influenced how individuals think about where and how they live. Especially in highly competitive metropolitan areas, issues such as affordability, housing accessibility, and socio-spatial inequality have become increasingly urgent. These challenges are not only economic concerns but also deeply linked to the broader sustainability agenda.

Within the framework of the United Nations Sustainable Development Goals (SDGs), Goal 11: Sustainable Cities and Communities explicitly calls for inclusive, safe, resilient, and sustainable urban environments. Housing lies at the very core of this vision. Affordable and adequate housing contributes to social equity, supports public health outcomes (SDG 3), and connects to infrastructure planning and innovation (SDG 9). In other words, housing functions as a convergence point where multiple SDGs intersect.

This project, therefore, positions housing as a strategic entry point for urban sustainability research. By applying data analytics techniques, the study aims to uncover patterns in pricing behavior, market heat, and regional disparities, providing insights that can inform urban planning and policymaking aligned with sustainability goals. The focus on housing allows for a localized, evidence-based exploration of sustainability challenges that are both measurable and socially impactful.

To guide this investigation, the following research questions were formulated:

- What are the key drivers of variation in housing prices across U.S. regions?
- How do affordability and market competitiveness indicators influence home sale prices?
- To what extent can machine learning models improve the prediction of housing prices compared to linear methods?
- What temporal trends can be observed in housing prices, and how might they evolve in the near future?

These questions provide the analytical foundation for the study and highlight the relevance of housing markets as a domain for sustainability-focused research.

Chapter 2

Exploratory Data Analysis

2.1 Data Preprocessing and Cleaning

2.1.1 Data Sources and Collection Methods

The dataset used in this project was obtained from **Zillow Research**, a public data platform that provides comprehensive and regularly updated information on the U.S. housing market. It includes a wide range of variables such as median sale prices, rental prices, market heat indices, affordability metrics, and new construction data. These indicators are essential for identifying temporal patterns and regional disparities in housing trends.

Although the initial intention was to conduct a comparative housing analysis using data from Bucharest, access to this data was restricted due to real estate privacy limitations. As a result, Zillow's datasets were selected as the primary source because of their high coverage, reliability, and time-based consistency.

The analysis focuses on the dynamics of urban housing markets, particularly how variables such as affordability and market demand fluctuate over time. Given the ongoing housing affordability crisis in many U.S. cities, understanding these patterns has become increasingly important for researchers, urban planners, and policymakers alike.

To enable structured time-series analysis, seven individual datasets were merged into a unified panel. These include:

- Median Sale Price
- Percentage of Homes Sold Above List Price
- Median Rent (ZORI)
- Market Heat Index
- Affordability (Years to Save)
- Sale-to-List Price Ratio

- New Construction Median Sale Price

Each dataset was reshaped into long format and merged using common identifiers: `RegionID`, `RegionName`, `RegionType`, `StateName`, and `Date` with using Python. The final dataset comprises 10,500 rows and 12 columns, covering the period from **March 2018 to February 2025**.
Variable Descriptions:

- **RegionID:** Unique code for each geographic region.
- **RegionName:** Name of the city or metro area.
- **RegionType:** Type of region.
- **StateName:** U.S. state where the region is located.
- **Date:** Monthly timestamp of the data.
- **Median Sale Price:** Typical home sale price in the region.
- **Pct Sold Above List:** Percentage of homes sold above listing price, indicating market competitiveness.
- **Median Rent:** Typical rental price across listed homes.
- **Market Heat Index:** Zillow's indicator of how competitive the housing market is.
- **YearsToSave:** Estimated time needed for a household to save for a standard down payment.
- **SaleToListRatio:** Ratio of sale price to original list price, showing pricing dynamics.
- **NewCon Median Sale Price:** Median price of newly constructed homes in the region.

2.1.2 Quality control procedures

To ensure the reliability and consistency of the dataset, a series of quality control procedures were applied. These included handling missing values, detection of outliers, removal of duplicate records, and verification of data types. Each step was implemented to preserve the integrity of the analysis pipeline and maintain consistency across all variables. Missing values and duplicate entries were identified using Python.

The initial dataset contained 10,500 rows and 12 variables. After merging multiple datasets, each row represented housing data for a specific city and month. However, not all cities had data available for every variable. For example, not every region includes newly constructed homes, which explains the missing values in the `NewConMedianSalePrice` column.

Smaller amounts of missing data in variables such as `PctSoldAboveList`, `SaleToListRatio`, and `MedianRent` were handled by removing only the affected rows. Basic data type checks confirmed appropriate formatting in Python. A heatmap was used to visualize missing values across columns. The most significant portion of missing data appeared in `NewConMedianSalePrice` (1,718 rows, approximately 16%), which was retained in order to preserve valuable information for regions where new construction activity is reported.

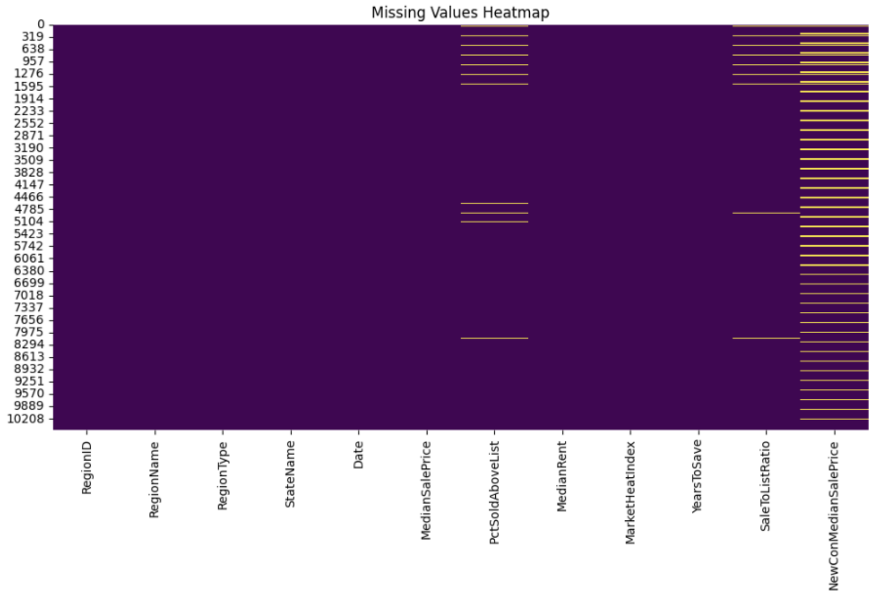
No duplicate records were detected. After the cleaning process, the dataset was reduced to 9,893 rows, ensuring consistency while maintaining as much valuable information as possible for further analysis.

```
Shape: (10500, 12)

 Column Types:
 RegionID                             int64
RegionName                          object
RegionType                          object
StateName                           object
Date                        datetime64[ns]
MedianSalePrice                    float64
PctSoldAboveList                   float64
MedianRent                         float64
MarketHeatIndex                    float64
YearsToSave                        float64
SaleToListRatio                    float64
NewConMedianSalePrice              float64
dtype: object

 Missing values per column:
 NewConMedianSalePrice    1718
PctSoldAboveList          301
SaleToListRatio           291
MedianRent                191
StateName                  84
dtype: int64
```

Figure 2.1: Data types and missing value summary for the merged dataset

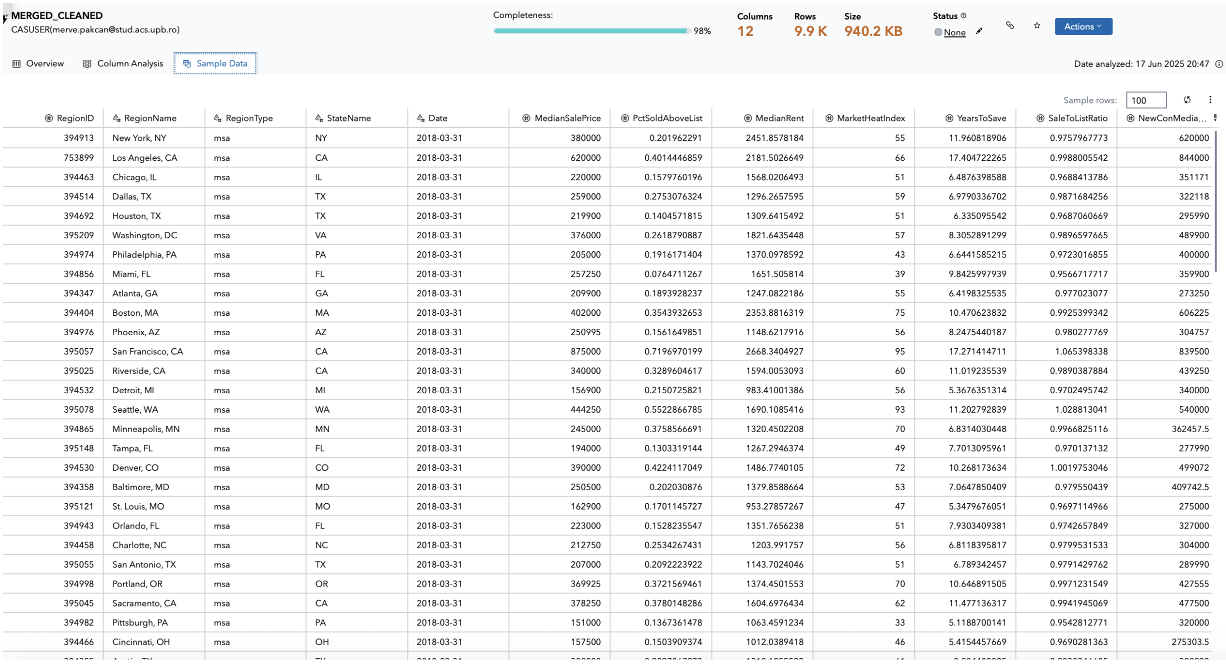



```
Rows before drop: 10500
Rows after drop (excluding only key variables): 9893

Duplicate Rows: 0
```

Figure 2.2: Missing values heatmap and post-cleaning row statistics

Following the merging and cleaning processes performed in Python, the dataset was imported into SAS via the Manage Data interface. Additionally, the Discover Information Assets part was utilized to explore summary statistics and gain a general overview of the dataset.The dependent variable used in this analysis is *MedianSalePrice*, representing housing market value. Independent variables include *MedianRent*, *NewConMedianSalePrice*, *PctSoldAboveList*, *SaleToListRatio*, *YearsToSave*, and *MarketHeatIndex*, as they are considered influential factors on home prices.



RegionID	RegionName	RegionType	StateName	Date	MedianSalePrice	PctSoldAboveList	MedianRent	MarketHeatIndex	YearsToSave	SaleToListRatio	NewConMedia...
394913	New York, NY	msa	NY	2018-03-31	380000	0.201962291	2451.8578184	55	11.960818906	0.9757967773	620000
753899	Los Angeles, CA	msa	CA	2018-03-31	620000	0.4014446859	2181.5026649	66	17.404722265	0.9988005542	844000
394463	Chicago, IL	msa	IL	2018-03-31	220000	0.1579760196	1568.0206493	51	6.4876398588	0.9688413786	351171
394514	Dallas, TX	msa	TX	2018-03-31	259000	0.2753076324	1296.2657595	59	6.9790336702	0.9871684256	322118
394692	Houston, TX	msa	TX	2018-03-31	219900	0.1404571815	1309.6415492	51	6.335095542	0.9687060669	295990
395209	Washington, DC	msa	VA	2018-03-31	376000	0.2618790887	1821.6435448	57	8.3052891299	0.9896597665	489900
394974	Philadelphia, PA	msa	PA	2018-03-31	205000	0.1916171404	1370.0978592	43	6.6441585215	0.9723016855	400000
394856	Miami, FL	msa	FL	2018-03-31	257250	0.0764711267	1651.505814	39	9.8425997939	0.9566717717	359900
394347	Atlanta, GA	msa	GA	2018-03-31	209900	0.1893928237	1247.0822186	55	6.4198325535	0.977023077	273250
394404	Boston, MA	msa	MA	2018-03-31	402000	0.3543932653	2353.8816319	75	10.470623832	0.9925399342	606225
394976	Phoenix, AZ	msa	AZ	2018-03-31	250995	0.1561649851	1148.6217916	56	8.2475440187	0.980277769	304757
395057	San Francisco, CA	msa	CA	2018-03-31	875000	0.7196970199	2668.3404927	95	17.271414711	1.065398338	839500
395025	Riverside, CA	msa	CA	2018-03-31	340000	0.3289604617	1594.0053093	60	11.019235539	0.9890387884	439250
394532	Detroit, MI	msa	MI	2018-03-31	156900	0.2150725821	983.41001386	56	5.3676351314	0.9702495742	340000
395078	Seattle, WA	msa	WA	2018-03-31	444250	0.5522866785	1690.1085416	93	11.202792839	1.028813041	540000
394865	Minneapolis, MN	msa	MN	2018-03-31	245000	0.3758566691	1320.4502208	70	6.8314030448	0.9966825116	362457.5
395148	Tampa, FL	msa	FL	2018-03-31	194000	0.1303319144	1267.2946374	49	7.7013095961	0.970137132	277990
394530	Denver, CO	msa	CO	2018-03-31	390000	0.4224117049	1486.7740105	72	10.268173634	1.0019753046	499072
394358	Baltimore, MD	msa	MD	2018-03-31	250500	0.202030876	1379.8588664	53	7.0647850409	0.979550439	409742.5
395121	St. Louis, MO	msa	MO	2018-03-31	162900	0.1701145727	953.27857267	47	5.3479676051	0.9697114966	275000
394943	Orlando, FL	msa	FL	2018-03-31	223000	0.1528235547	1351.7656238	51	7.9303409381	0.9742657849	327000
394458	Charlotte, NC	msa	NC	2018-03-31	212750	0.2534267431	1203.991757	56	6.8118395817	0.9799531533	304000
395055	San Antonio, TX	msa	TX	2018-03-31	207000	0.2092223922	1143.7024046	51	6.789342457	0.9791429762	289990
394998	Portland, OR	msa	OR	2018-03-31	369925	0.3721569461	1374.4501553	70	10.646891505	0.9971231549	427555
395045	Sacramento, CA	msa	CA	2018-03-31	378250	0.3780148286	1604.6976434	62	11.477136317	0.9941945069	477500
394982	Pittsburgh, PA	msa	PA	2018-03-31	151000	0.1367361478	1063.4591234	33	5.1188700141	0.9542812771	320000
394466	Cincinnati, OH	msa	OH	2018-03-31	157500	0.1503909374	1012.0389418	46	5.4154457669	0.9690281363	275303.5

Figure 2.4: Sas Discover Information Assets

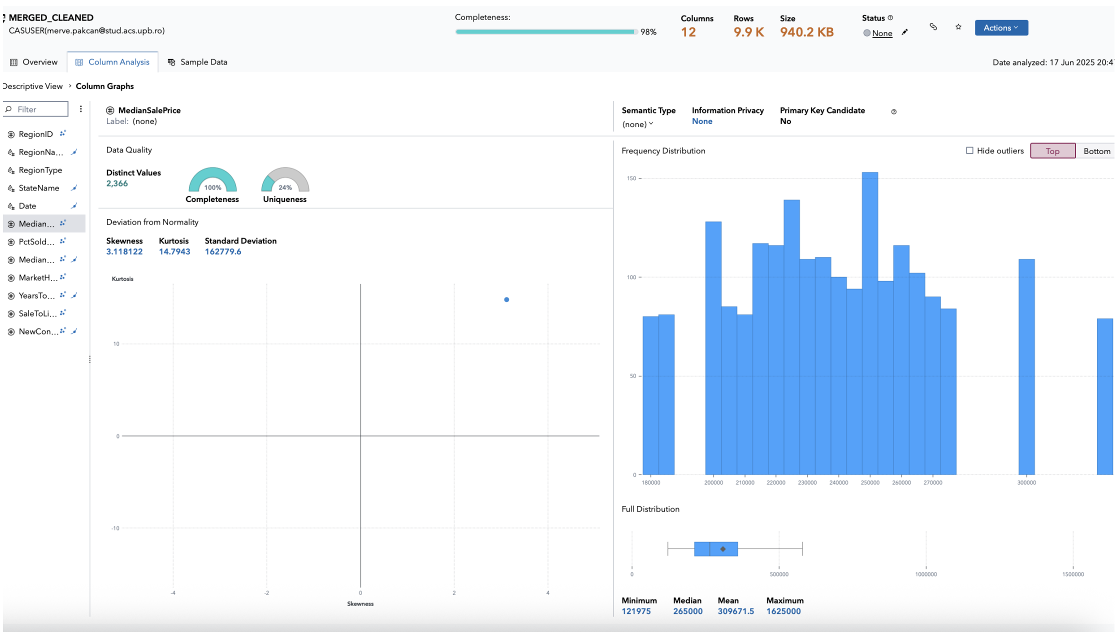


Figure 2.3: Column analysis of the cleaned and merged Zillow dataset

The dataset comprises 9,900 rows and 12 columns, with a total size of 940.2 KB.

Out of these 12 columns, 8 are numerical (DataType: double) and 4 are character data (DataType: varchar), offering a robust data structure for analysis.

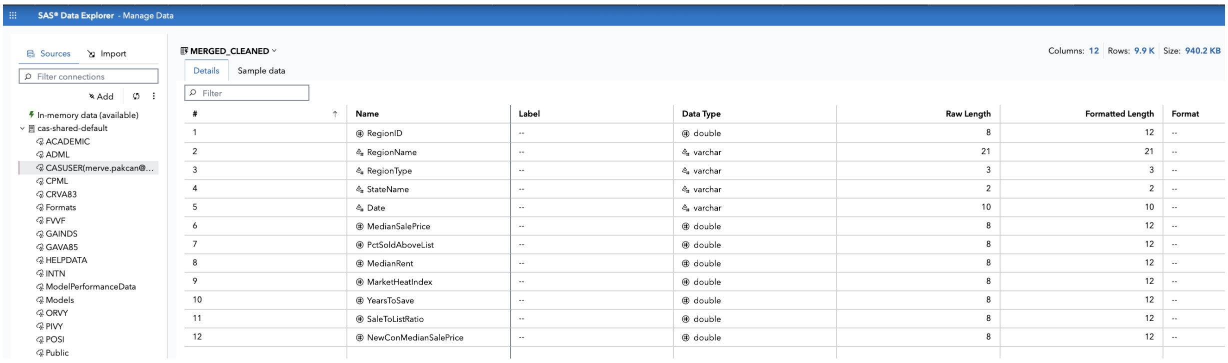


Figure 2.5: Data type summary of the cleaned dataset in SAS Data Explorer

Also, after loaded the data into SAS environment, data type controlled in Manage Data part.

Outlier detection was performed using the Explore and Visualize tools in SAS. Box plots were generated for key numerical variables to examine the distribution and identify any extreme values. Rather than displaying individual data points, SAS visualizations use gradient-shaded areas to indicate regions where outliers are concentrated. This approach provides a clear understanding of variability while maintaining a clean and interpretable visual layout.

Figure 2.6: Box plot for `MedianSalePrice`

The variables *MedianSalePrice* and *NewConMedianSalePrice* exhibited right-skewed distributions with clearly defined upper outlier zones. These outliers were particularly prominent in high-cost urban regions, reflecting actual market conditions rather than anomalies.

PctSoldAboveList and *SaleToListRatio* also showed high-value outliers, consistent with competitive market dynamics where homes frequently sell above their list prices. These patterns are particularly visible through the shaded bands above the box range.

In the case of *YearsToSave* and *MarketHeatIndex*, outliers indicated regions where affordability and buyer competition significantly diverged from the national median. These values were not removed, as they represent meaningful variations in housing market behavior.

Retaining such outliers was considered essential, as they capture important aspects of real-world market diversity and contribute to the explanatory power of the dataset in subsequent analyses.

Exploratory Data Analysis

2.1.3 Descriptive Statistics

	RegionID	Date	MedianSalePrice
count	9893.000000	9893	9.893000e+03
mean	406601.992621	2021-10-16 19:09:28.038006528	3.096715e+05
min	394312.000000	2018-03-31 00:00:00	1.219750e+05
25%	394531.000000	2020-01-31 00:00:00	2.131770e+05
50%	394792.000000	2021-10-31 00:00:00	2.650000e+05
75%	395005.000000	2023-07-31 00:00:00	3.599000e+05
max	753912.000000	2025-02-28 00:00:00	1.625000e+06
std	64112.991749	NaN	1.627796e+05

	PctSoldAboveList	MedianRent	MarketHeatIndex	YearsToSave
count	9893.000000	9893.000000	9893.000000	9893.000000
mean	0.306217	1500.371539	61.627211	8.631004
min	0.029102	806.913837	17.000000	4.670567
25%	0.175565	1208.192870	50.000000	6.822396
50%	0.271981	1402.893622	59.000000	7.959976
75%	0.416761	1715.942181	71.000000	9.836859
max	0.859512	3347.407630	197.000000	22.269278
std	0.163279	429.354211	17.170322	2.625767

	SaleToListRatio	NewConMedianSalePrice
count	9893.000000	8.481000e+03
mean	0.988815	4.059965e+05
min	0.943317	1.620000e+05
25%	0.975242	3.062750e+05
50%	0.985311	3.650000e+05
75%	0.998708	4.499900e+05
max	1.140964	2.510000e+06
std	0.020732	1.759764e+05

Figure 2.7: Summary statistics of the key variables in the Zillow dataset

Table 2.7 provides a statistical overview of the main variables used in the analysis. The `MedianSalePrice` shows substantial variation across regions, with a mean of approximately $309,000 and a maximum value exceeding $1.6 million. New construction properties (`NewConMedianSalePrice`) reach up to $2.5 million in some cities, indicating high-end market segments.

The `PctSoldAboveList` variable, with an average of 36%, suggests that a significant portion of homes are sold above the listing price—an indication of competitive housing markets in certain regions.

Rental prices (`MedianRent`) range from $806 to over $3,300, while the `MarketHeatIndex` spans from 17 to 197, reinforcing the presence of stark regional differences in demand and buyer activity.

Notably, the `YearsToSave` variable, which estimates the number of years required to save for a down payment, averages at 8.6 years and reaches beyond 22 years in some areas. This underscores significant affordability challenges across cities.

These preliminary statistics offer key insights into housing price dynamics, regional competitiveness, and financial accessibility, setting a solid foundation for further exploratory analysis.

2.1.4 Distribution Analysis via Histograms

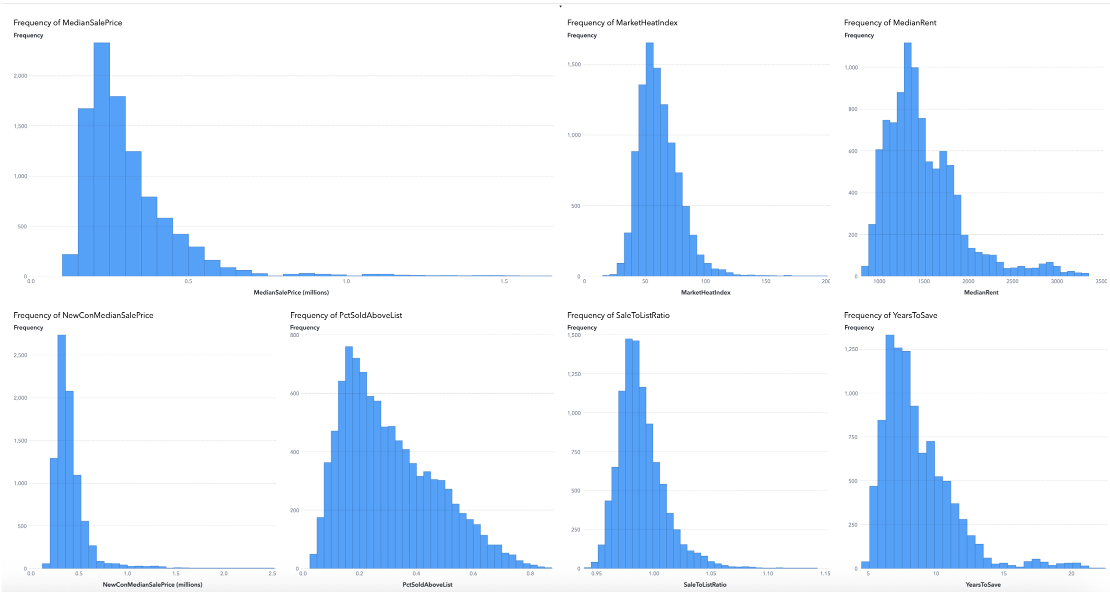


Figure 2.8: Distribution of key Zillow housing indicators.

The distribution of key numerical variables was examined using histograms generated in the SAS Visual Analytics environment. These plots provided a clear visual summary of the frequency and skewness patterns across the dataset.

Variables such as *MedianSalePrice*, *NewConMedianSalePrice*, and *MedianRent* demonstrated strong right-skewed distributions. This is consistent with real-world housing markets, where a majority of properties cluster around mid-range values, but a smaller portion of high-value listings extend the upper tail. These skewed patterns highlight the price diversity and economic inequality present in different housing regions.

Conversely, *MarketHeatIndex* followed a near-normal distribution, indicating that most markets experience moderate buyer competition, with fewer cases showing exceptionally high or low heat levels. *PctSoldAboveList* and *SaleToListRatio* also revealed slight right skewness, supporting the presence of aggressive market conditions where homes often sell above their listing price.

YearsToSave showed a right-skewed profile as well, with the majority of regions requiring a moderate savings period, while a smaller number of areas—typically high-cost urban centers—demand significantly longer timeframes to afford a median-priced home.

Overall, the histograms validate the presence of non-normality and economic variability within the dataset. These characteristics provide valuable context for model development and interpretation in subsequent stages of the analysis.

2.1.5 Correlation Analysis of Housing Indicators

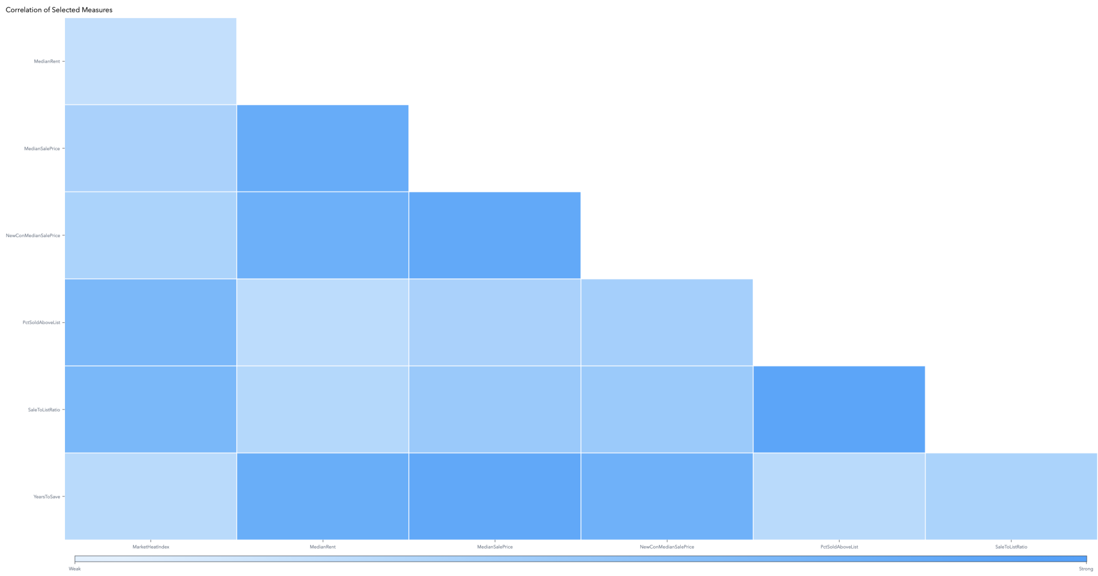


Figure 2.9: Correlation matrix

The correlation matrix illustrates the strength and direction of linear relationships among selected housing variables. As expected, strong positive correlations are observed between *MedianSalePrice*, *MedianRent*, and *YearsToSave*, suggesting that higher home prices are generally associated with higher rental costs and longer saving periods. Similarly, the strong association between *PctSoldAboveList* and *SaleToListRatio* reflects market competitiveness in overheated regions. In contrast, variables such as *MarketHeatIndex* show more moderate correlations, indicating their distinct but complementary role in explaining housing dynamics.

2.1.6 Q-Q Plots for Normality Assessment

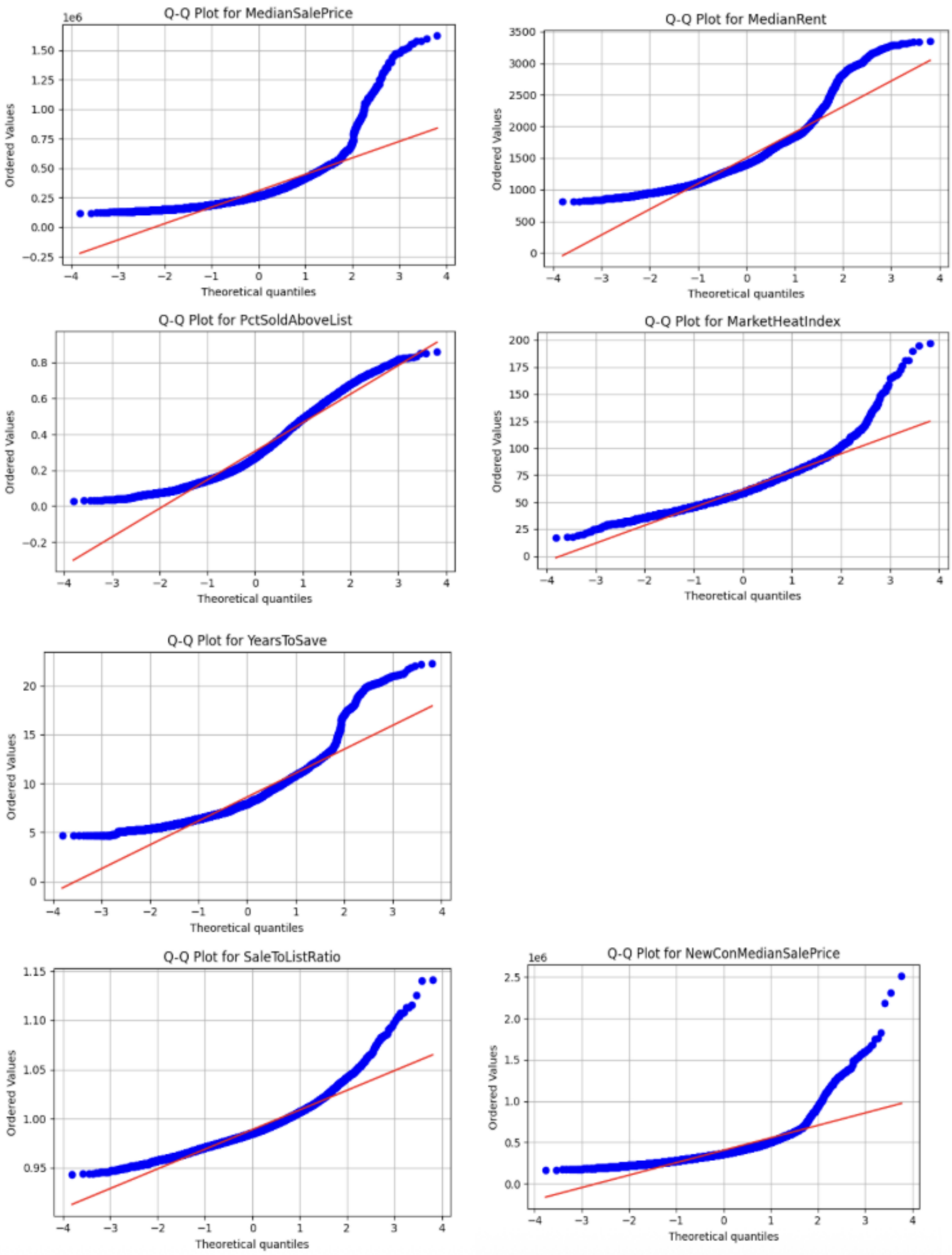


Figure 2.10: Q-Q plots

Q–Q plots were used to assess the normality of key numerical variables in the housing dataset. Most variables, including `MedianSalePrice`, `MedianRent`, and `YearsToSave`, display clear deviations from the theoretical reference line, particularly at the tails. This indicates strong right-skewness and the presence of extreme values.

For example, `MedianSalePrice` and `MedianRent` exhibit pronounced curvature at both ends of the distribution, reflecting a small number of regions with exceptionally high prices. Similarly, variables such as `PctSoldAboveList` and `MarketHeatIndex` show S-shaped or curved patterns, suggesting departures from normality and potential clustering.

These distributional patterns confirm that the normality assumption does not hold for most features. Therefore, non-parametric or distribution-agnostic methods are adopted in the following analysis to ensure statistical robustness and valid inference.

2.1.7 Time Series Analysis of Rent and Sale Prices

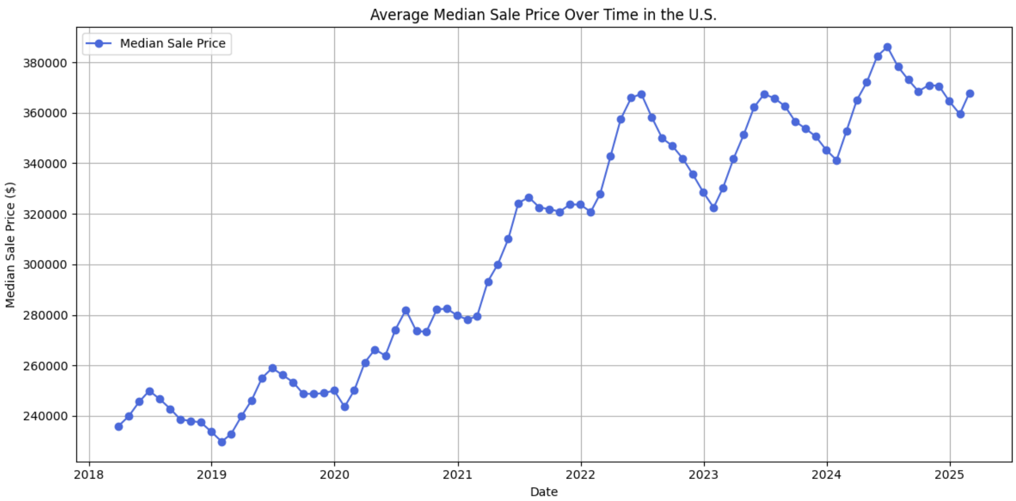


Figure 2.11: Time series plots of average monthly Median Sale Price

To capture long-term trends in housing dynamics, time series plots were generated for average monthly values of median sale price and median rent in the U.S. between 2018 and early 2025. Median Sale Price shows a steady upward trend with minor fluctuations. A significant acceleration is observed around 2021–2022, potentially reflecting post-pandemic housing demand surges and low interest rates during that period.

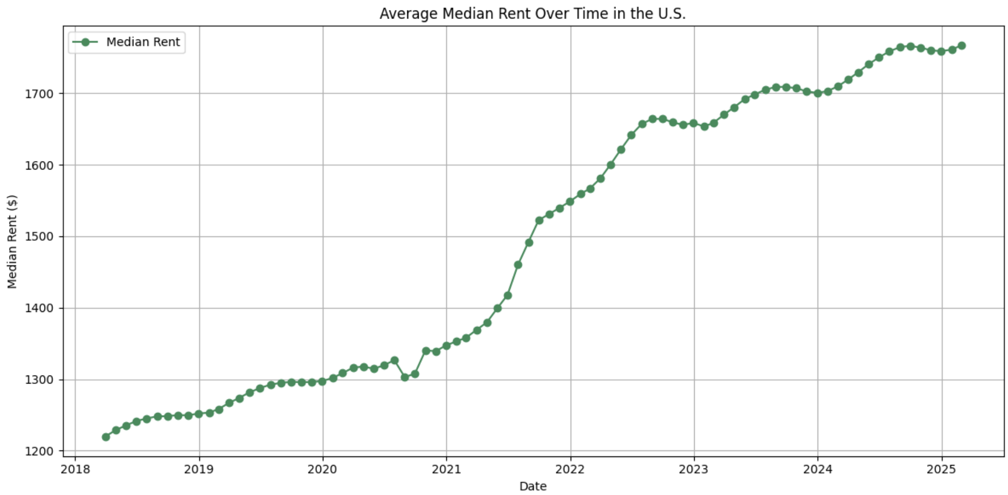


Figure 2.12: Time series plots of average monthly Median Rent Price

Median Rent follows a similar pattern, increasing gradually before rising more sharply from late 2021 onward. This may indicate increasing rental pressure in parallel with rising home prices. These temporal patterns suggest a strong cyclical growth in housing costs over recent years, reinforcing concerns around affordability and supporting the need for sustainable urban planning approaches.

2.1.8 Regional Analysis

To better illustrate the substantial differences between median sale prices and rents across states, the data was sorted in descending order. This highlights the regional disparities more clearly.

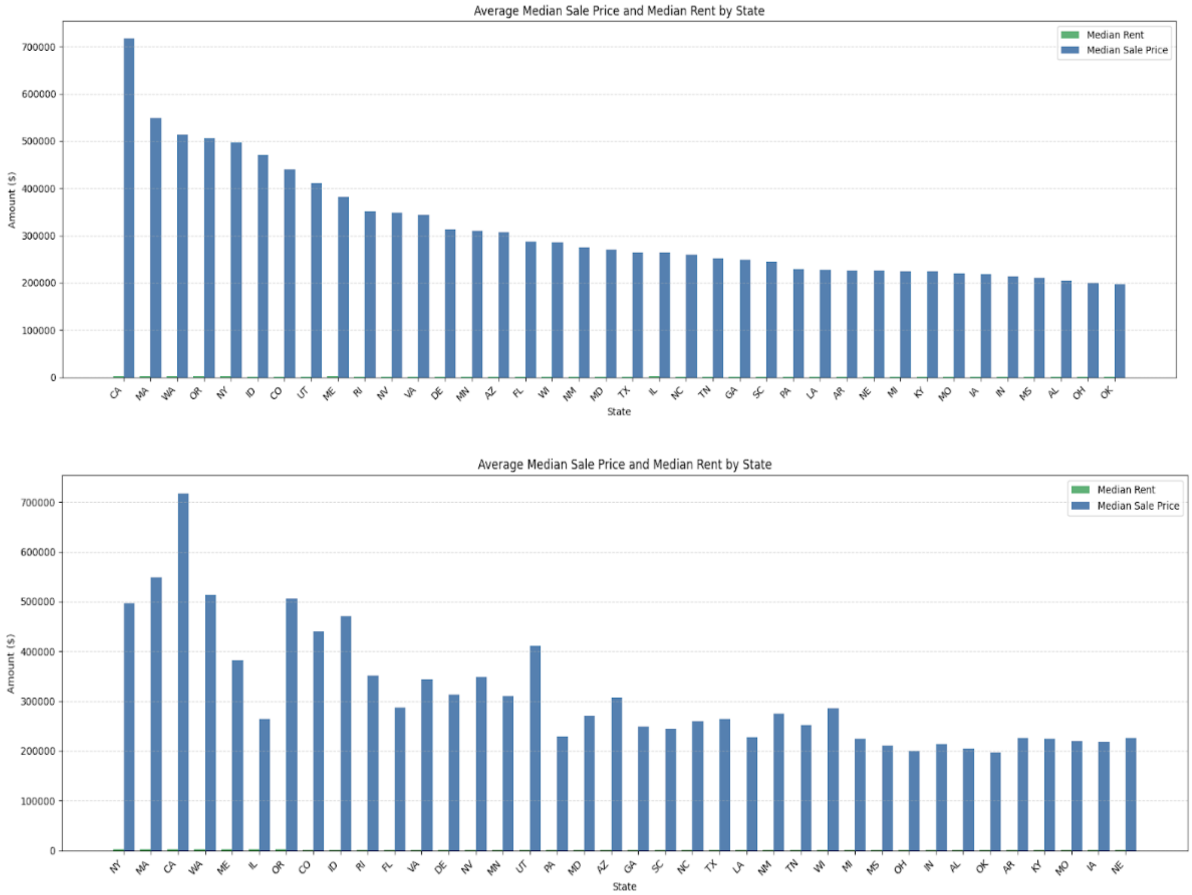


Figure 2.13: Comparison of average median sale price and median rent across U.S. states.

California has the highest median home sale price, followed by Washington and Massachusetts, all of which also report relatively high rental costs. Notably, New York has the highest median rent but does not appear among the top three states for sale prices.

In states like Idaho and Utah, home prices are comparatively high while rents remain moderate, resulting in a significant gap between ownership and rental markets. In contrast, Southern states such as Oklahoma and Mississippi exhibit both low sale prices and low rents, reflecting more affordable housing conditions.

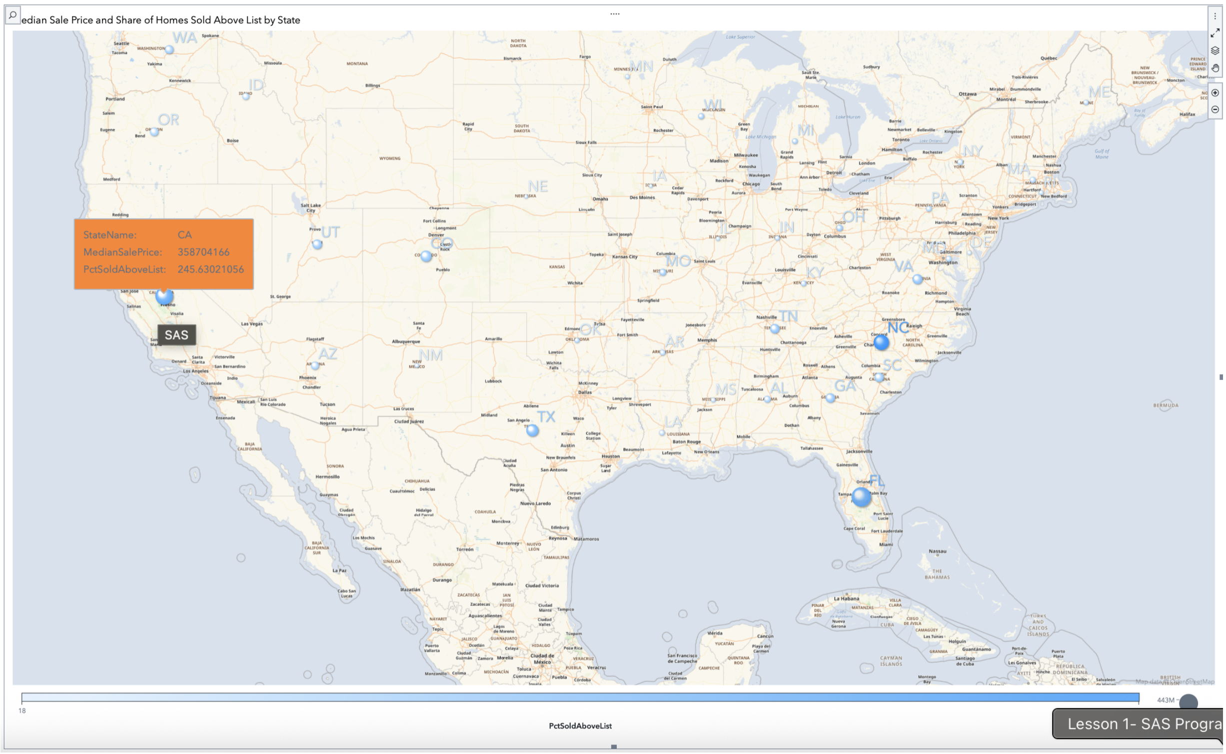


Figure 2.14: Geo Bubble Map showing Median Sale Price (size) and `PctSoldAboveList` (color) across U.S. states

Geographic Distribution of Median Sale Price and Market Competitiveness

The Geo Bubble Map provides a spatial overview of the U.S. housing market by visualizing the Median Sale Price (bubble size) and the PctSoldAboveList ratio (color intensity) across states. Larger bubbles indicate higher average sale prices, while more saturated blue tones reflect greater market competitiveness, a higher share of homes sold above the listing price. States like California (CA) and Florida (FL) exhibit both high prices and intense buyer competition, highlighting highly active housing markets. This visualization supports regional comparison and reveals that price and competitiveness often concentrate in coastal and metropolitan areas.

Chapter 3

Feature Engineering and Model Design

To enhance the analytical depth of this study, three derived features were engineered from the original Zillow housing dataset using SAS Visual Analytics:

- **RentToPriceRatio:** Calculated as `MedianRent` divided by `MedianSalePrice`, this metric serves as a proxy for affordability and rental yield. It reflects the relative cost of renting versus owning a home.
- **MarketTension:** Constructed by multiplying `MarketHeatIndex` with `PctSoldAboveList`, this composite indicator captures latent demand and market competitiveness, highlighting regions with stronger upward pricing pressures.
- **LogMedianSalePrice:** This is the logarithm of `MedianSalePrice`. It is used to reduce skewness and stabilize variance, thereby improving model interpretability and performance in regression-based analyses.

These engineered features aim to better capture hidden market dynamics and are later used in exploratory analyses and predictive modeling.

To examine the factors influencing housing prices, a series of predictive models were constructed using SAS Visual Analytics. The aim was to evaluate how key indicators affect home sale prices across different U.S. regions.

The modeling process began with linear regression to establish a baseline and uncover interpretable relationships. This was followed by tree-based methods, including decision trees and ensemble models, to better capture non-linear interactions within the data.

Applying multiple modeling techniques enabled a comprehensive comparison of predictive accuracy, assessment of variable importance, and validation of results across approaches. Throughout the analysis, `LogMedianSalePrice` was used as the target variable to reduce skewness and improve both model fit and interpretability.

3.0.1 Linear Regression

Linear regression was used as a baseline model to understand the direct relationships between predictors and the target variable.

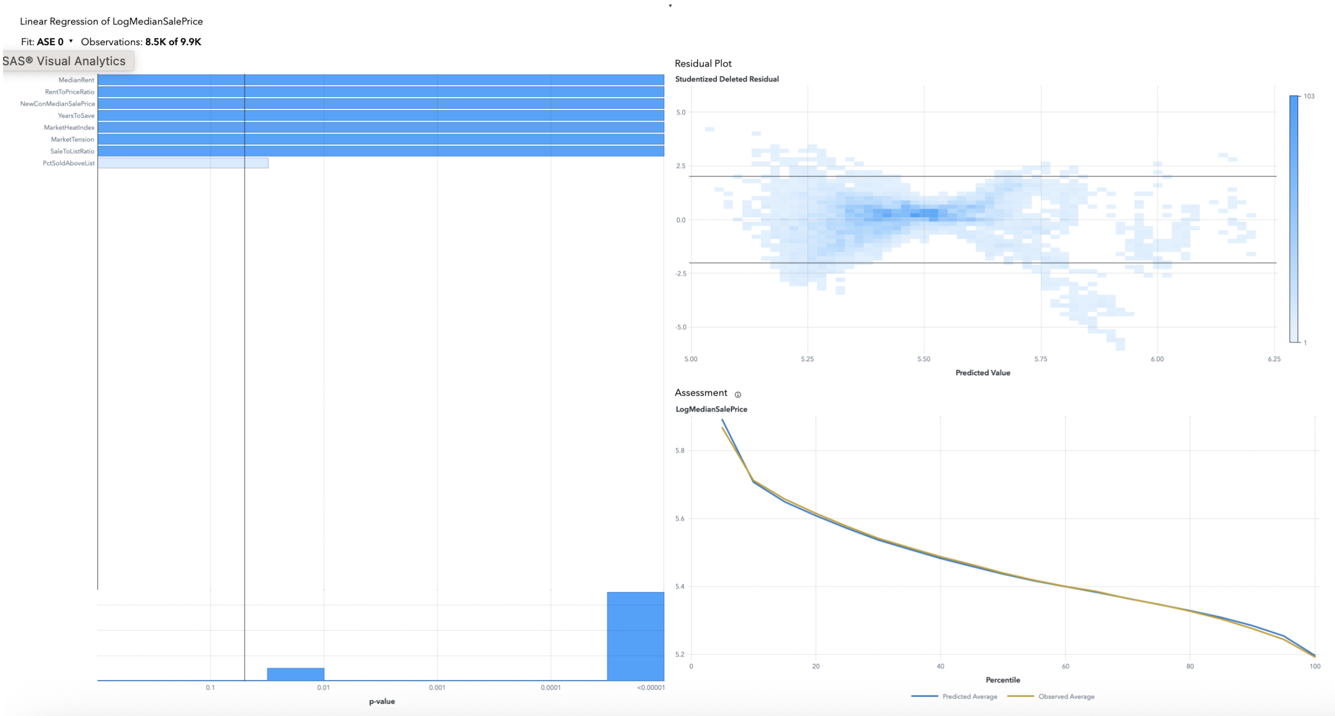


Figure 3.1: Linear regression model

The linear regression model indicated statistically significant associations between the transformed target variable, key predictors such as `MedianRent`, `RentToPriceRatio`, `NewConMedianSalePrice`, and `YearsToSave`. The assessment curve showed strong alignment between predicted and observed values across percentiles, supporting the model's validity as a baseline approach.

However, the residual plot suggested non-linearity and signs of heteroscedasticity, with residuals spreading unevenly across predicted values. These limitations highlighted the linear model's inability to fully capture market complexity. As a result, more adaptive, non-linear models were pursued in subsequent analysis stages.

3.0.2 Random Forest

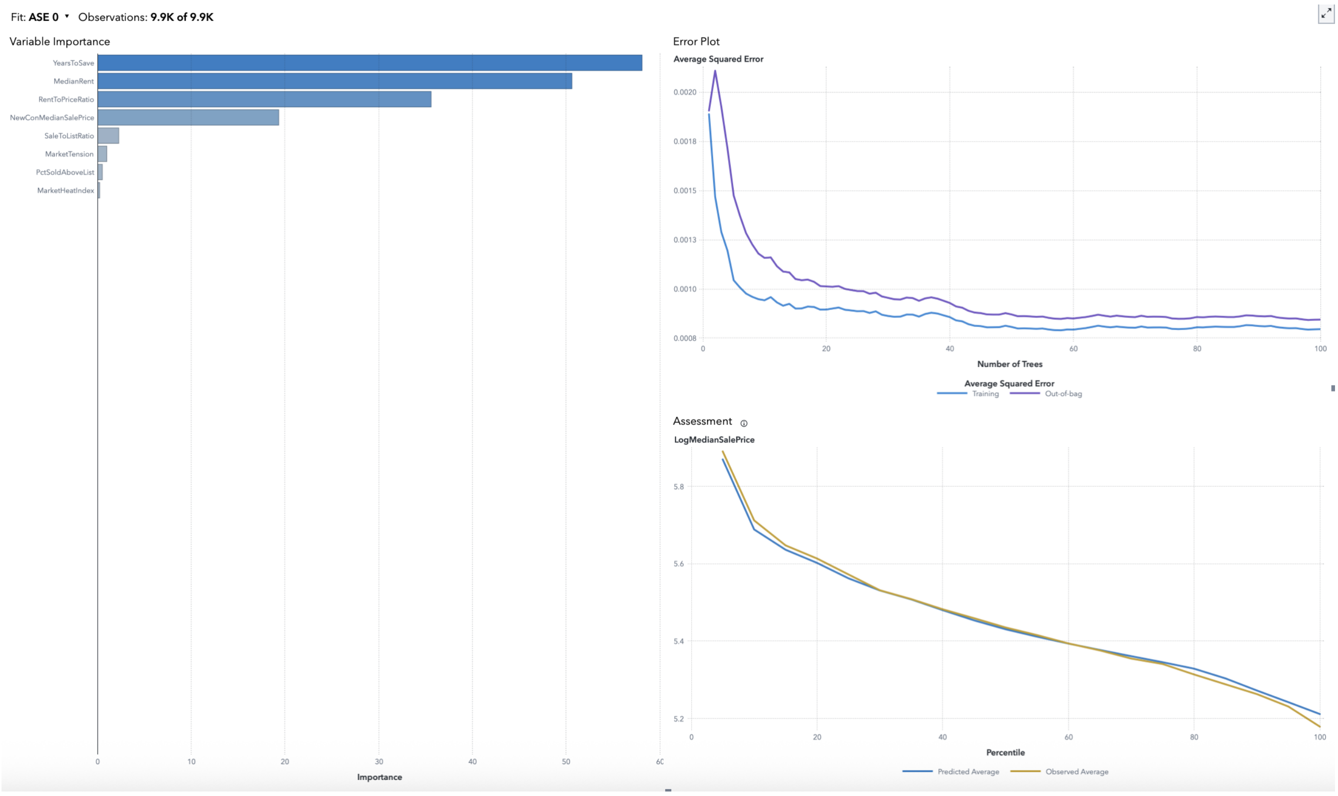


Figure 3.2: Random forest model

The random forest model demonstrated strong predictive accuracy for `LogMedianSalePrice`, with low out-of-bag error and strong agreement between predicted and observed values across all percentiles. The error plot confirmed that training and validation errors converged steadily, suggesting a stable and generalizable model with low risk of overfitting.

`YearsToSave`, `MedianRent`, and `RentToPriceRatio` emerged as the top contributing features in the variable importance chart. Compared to the linear model, the random forest provided enhanced flexibility and captured more complex relationships, improving overall predictive power and robustness.

3.0.3 Decision Tree

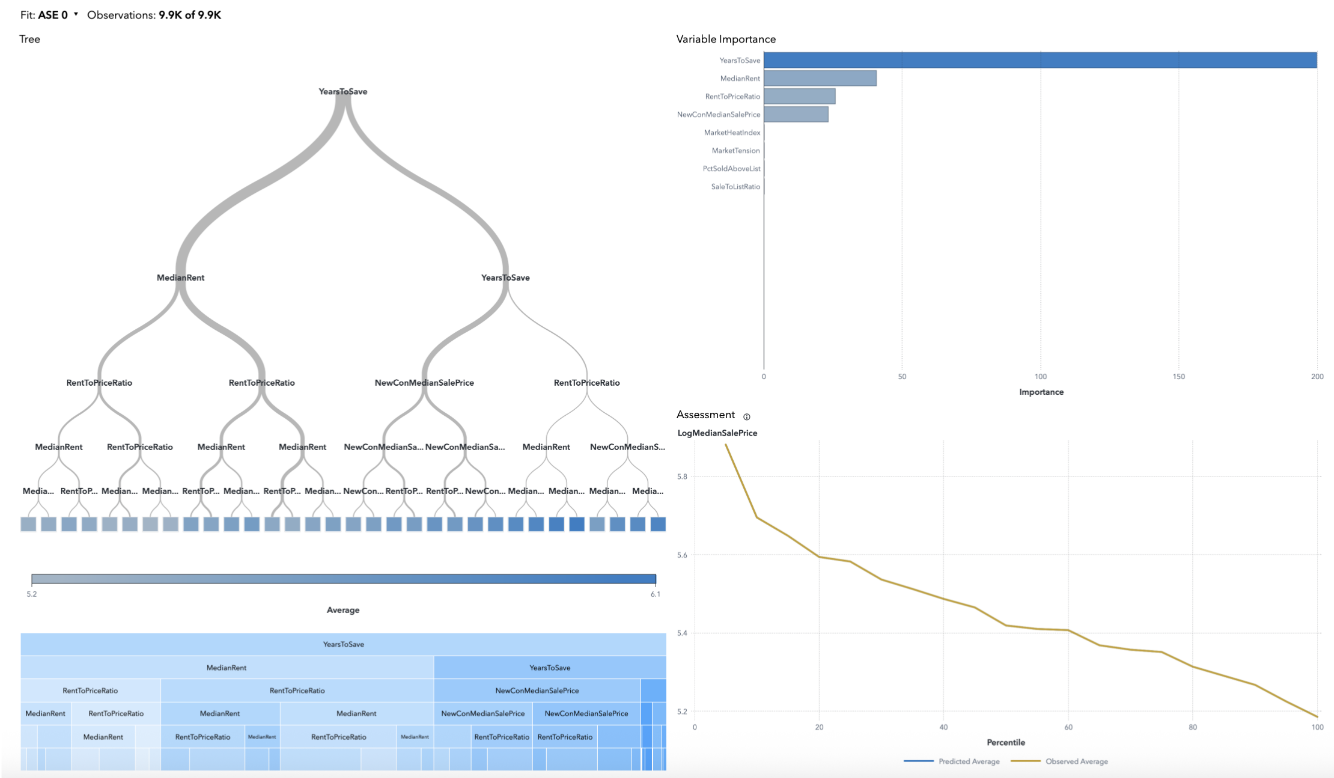


Figure 3.3: Decision tree model

The decision tree model showed that `YearsToSave` was the most important variable for predicting `LogMedianSalePrice`, followed by `MedianRent` and `RentToPriceRatio`. The tree made clear splits based on affordability, making the structure easy to follow and interpret.

Even though the model picked up on meaningful patterns, the assessment curve pointed to some underfitting, likely due to the limits of using a single tree. Still, it helped reveal how the variables relate to each other and offered a good view of their overall influence. .

3.0.4 Gradient Boosting

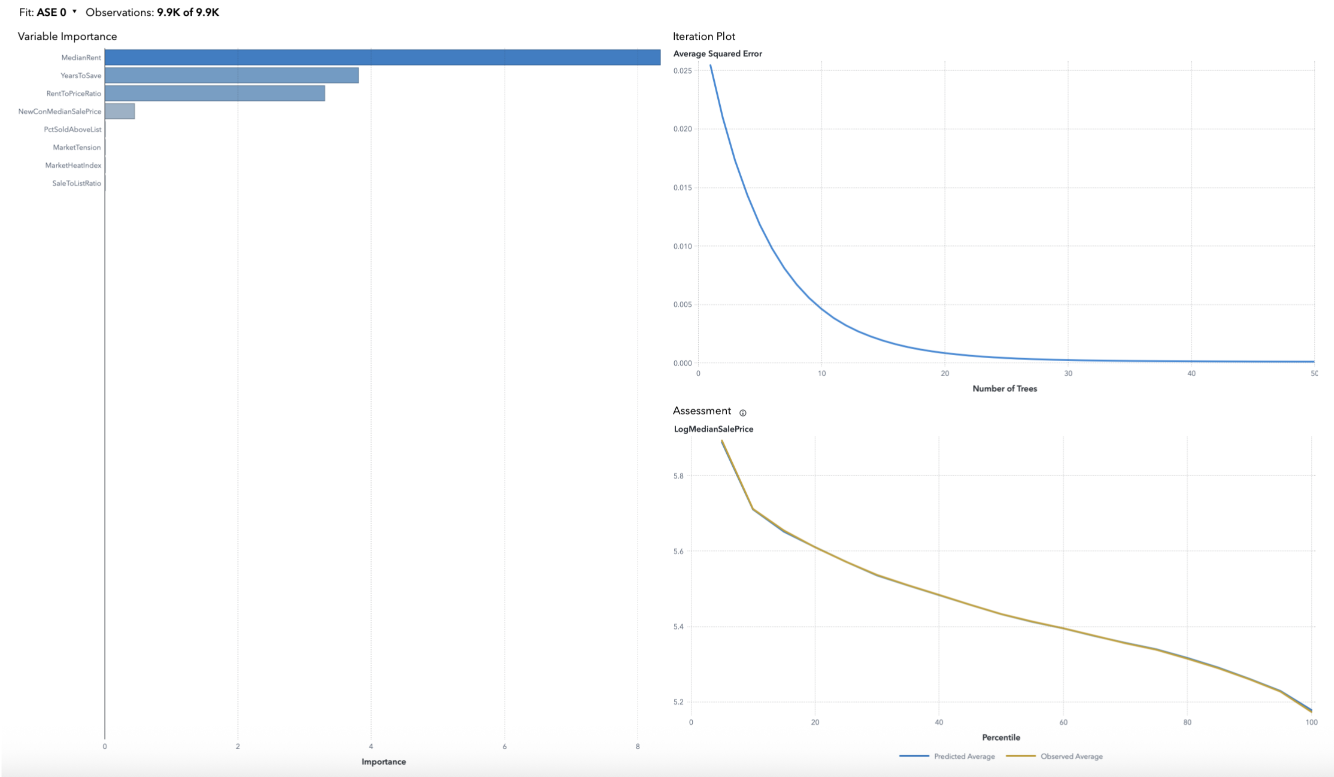


Figure 3.4: Gradient boosting model

The gradient boosting model showed strong accuracy. The error dropped quickly and stabilized early. The most influential predictors were `MedianRent`, `YearsToSave`, and `RentToPriceRatio`. Other variables had minimal impact, which could be due to feature overlap or reduced non-linear effects after log transformation. The predicted values aligned closely with the observed values across percentiles.

3.0.5 Model Comparison

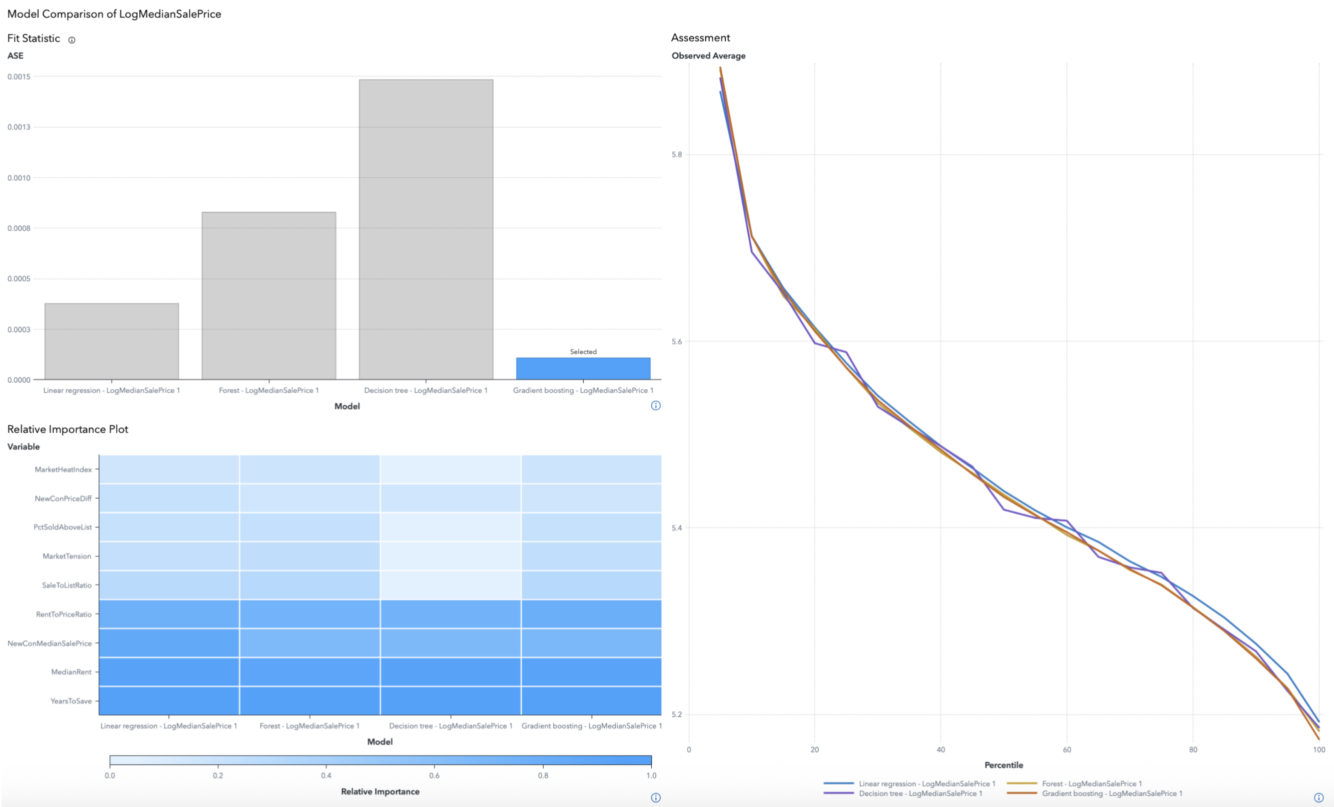


Figure 3.5: Model comparison

Among the evaluated models, **Gradient Boosting** achieved the lowest ASE, confirming its superior predictive accuracy. **Random Forest** followed closely, while **Linear Regression** provided a solid baseline. The **Decision Tree**, although interpretable, showed the weakest performance.

Assessment curves showed that Gradient Boosting most closely aligned with the observed values, especially across middle and upper percentiles, reflecting robust generalization.

Across all models, `YearsToSave` and `MedianRent` consistently ranked among the top predictors. This reinforces their strong influence on `LogMedianSalePrice` and highlights their importance in housing market dynamics.

Chapter 4

Future Direction

4.0.1 Future Price Projection Using Time Series Analysis

To add a future-looking element to the project, a time series forecast was made using the Prophet library in Python. The goal was to take a first step toward including time-based trends in housing price analysis. Using data from 2018 to 2025, the model predicted median sale prices for the next two years.

As it seen from the graph, the forecast shows a steady upward trend, though the confidence interval gets wider over time. This suggests more uncertainty further out. Still, the general direction fits with the pattern that is seen in the past data. While this was just a basic attempt, it shows that time series models can be useful alongside other methods to better understand housing market trends.The deep analysis will be done

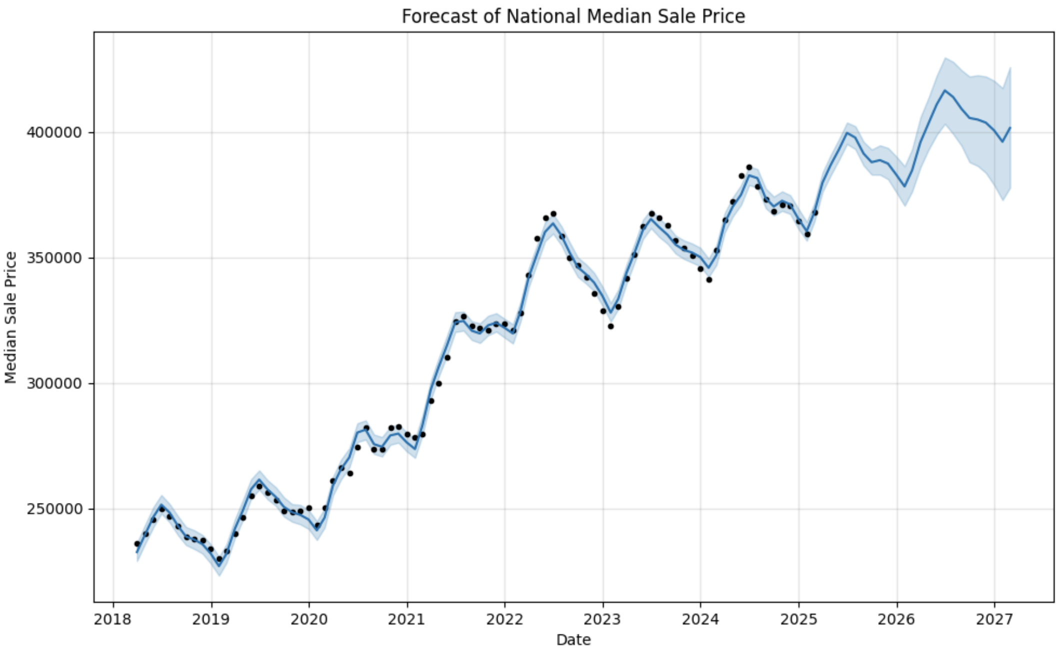


Figure 4.1: Prophet

Chapter 5

Conclusion

This project examined the main factors influencing housing prices in U.S. cities by combining data preparation, exploratory analysis, and predictive modeling. Using Zillow's dataset covering the years 2018 to 2025, the project focused on how variables related to affordability, rental pressure, and market activity shape price levels across regions.

Exploratory data analysis helped uncover initial patterns through distribution plots, summary statistics, and correlation heatmaps. These steps highlighted key relationships and regional differences, which guided the selection of features for modeling.

Linear regression was used as a starting point, followed by decision trees, random forests, and gradient boosting to capture more complex patterns. Among all, gradient boosting showed the best predictive performance. Across models, `YearsToSave` and `MedianRent` consistently stood out as the most important features.

To support a forward-looking perspective, a simple forecast using the Prophet model in Python was added. This time series projection extended the analysis by showing expected future price trends based on recent historical data.

Overall, the project shows how different data science techniques can be used together to understand urban housing dynamics. The results emphasize the role of affordability and competition in shaping housing prices and provide a strong basis for future work in real estate analysis and urban policy.

Appendix A

Fișiere sursă

[language=matlab,caption=Cod Matlab – fișier complet,label=lst:s1]./code/s1.m [language=matlab,caption=Cod Matlab – fragment de fișier,label=lst:s2,firstline=5,lastline=15]./code/s2.m